Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Kubernetes Fundamentals for Beginners

Course Code: C742

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	2 October 2026	Initial release — Kubernetes Fundamentals for Beginners (C742) lesson plan covering all 16 labs across five topics.	Dr. Alfred Ang

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Course Information

Course Title	Kubernetes Fundamentals for Beginners
Course Code	C742
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 7.5 instructional hours per day (15 hours)
Daily Timing	9:30 am – 5:30 pm (30-minute lunch)
Mode	Instructor-led, hands-on Kubernetes labs on a local minikube cluster
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Develop an architectural proof of concept for Kubernetes architecture and components.
- LO2: Develop solution architecture by applying container orchestration tools, techniques, and system modelling methods.
- LO3: Identify technical requirements and prepare a technical blueprint for Cloud Native Architecture.
- LO4: Demonstrate how Cloud Native Observability solutions and components address a specific business problem effectively.
- LO5: Implement system reviews to monitor solution status and adapt according to Cloud Native frameworks.

Course Schedule

Day 1 — Kubernetes Fundamentals & Container Orchestration

Time	Duration	Topic / Activity	Slides
9:30–9:45	15 min	Welcome, course introduction, ground rules and learning outcomes	1-10
9:45–10:25	40 min	Topic 1 — Kubernetes Fundamentals and Architecture: why Kubernetes, VMs vs containers, the reconciliation loop, declarative vs imperative, control-plane and node components	45-70

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		(concepts + demo)	
10:25–12:05	100 min	Hands-on: Lab 1: Bootstrap a Kubernetes Cluster and Explore the Architecture; Lab 2: Work with the Kubernetes API and kubectl	49-62
12:05–13:05	60 min	Topic 1 continued — workload resources (Pod, ReplicaSet, Deployment, StatefulSet, DaemonSet, Job/CronJob), labels and selectors, multi-container Pod patterns (concepts + demo)	45-70
13:05–13:35	30 min	Lunch break	—
13:35–14:55	80 min	Hands-on: Lab 3: Deploy and Manage Pods, ReplicaSets and Deployments; Lab 4: Control Pod Scheduling with Requests, Taints and Affinity	63-76
14:55–15:45	50 min	Topic 2 — Container Orchestration and Kubernetes Operations: containers and the OCI standard, CRI runtimes, the Kubernetes network model, Services and Ingress (concepts + demo)	78-103
15:45–17:15	90 min	Hands-on: Lab 5: Build and Run Containers with Docker and the OCI Standard; Lab	82-95

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		6: Expose Applications with Services and Ingress	
17:15–17:30	15 min	Day 1 recap and Q&A	—

Total instructional time: 450 minutes (7.5 hours), 9:30 am – 5:30 pm with a 30-minute lunch.

Day 2 — Security, Cloud Native Architecture, Observability & Delivery

Time	Duration	Topic / Activity	Slides
9:30–9:45	15 min	Day 1 recap	—
9:45–11:00	75 min	Topic 2 continued — the 4C security model, RBAC, ServiceAccounts, Secrets, NetworkPolicy, securityContext, storage (PV/PVC/StorageClass/CSI) and service mesh (concepts + demo)	78-103
11:00–12:40	100 min	Hands-on: Lab 7: Secure the Cluster with RBAC, Secrets and NetworkPolicy; Lab 8: Persist Data with PersistentVolumes, StatefulSets and Storage Classes; Lab 9: Autoscale Workloads with the Horizontal Pod Autoscaler	96-121
12:40–13:40	60 min	Topic 3 — Cloud Native Architecture and Scalability: autoscaling (HPA/VPA/Cluster Autoscaler), resilience patterns, serverless, open standards and CNCF governance (concepts). Hands-on: Lab 10: Build Resilience with Probes, Disruption	122-128

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		Budgets and Multi-Tier Architecture	
13:40–14:10	30 min	Lunch break	—
14:10–15:10	60 min	Hands-on: Lab 11: Explore Serverless, Open Standards and the CNCF Landscape	129-135
15:10–16:00	50 min	Topic 4 — Kubernetes Monitoring and Observability: the three pillars, Prometheus and PromQL, the four golden signals, symptom-based alerting and cost management (concepts + demo)	137-148
16:00–16:50	50 min	Hands-on: Lab 12: Instrument and Collect Metrics with Prometheus; Lab 13: Apply the Golden Signals, Alerting and Cost Management	141-154
16:50–17:15	25 min	Topic 5 — Kubernetes Application Deployment and Management: rollouts and progressive delivery, Helm and Kustomize, GitOps and CI/CD (concepts). Hands-on: Lab 14: Perform Zero-Downtime Rolling Updates and Canary Releases; Lab 15: Package Applications with Helm and Kustomize; Lab 16: Automate Delivery with GitOps and	160-180

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		CI/CD, and Conduct a System Review	
17:15–17:30	15 min	Course recap, Q&A and course feedback	182-187

Total instructional time: 450 minutes (7.5 hours), 9:30 am – 5:30 pm with a 30-minute lunch.

Lab Reference (aligned to exam skill areas)

Topic / Exam skill area	Weighting	Labs
Topic 01: Kubernetes Fundamentals and Architecture	46%	Lab 1, Lab 2, Lab 3, Lab 4
Topic 02: Container Orchestration and Kubernetes Operations	22%	Lab 5, Lab 6, Lab 7, Lab 8
Topic 03: Cloud Native Architecture and Scalability	16%	Lab 9, Lab 10, Lab 11
Topic 04: Kubernetes Monitoring and Observability	8%	Lab 12, Lab 13
Topic 05: Kubernetes Application Deployment and Management	8%	Lab 14, Lab 15, Lab 16

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